

3. Dik: HIMPUNAN A, B, C sebagai berikut: $A = \{16, 18, 20, 23, 26, 29, 31\}$
 $B = \{19, 21, 23, 26, 28, 34\}$
 $C = \{17, 23, 26, 27, 33, 36\}$
 $U = \{16, 17, 18, \dots, 36\}$

a. Hitung Manual

1. $A \cap B \cap C = \{23, 26\} = 2$

2. $A \cup B \cup C = \{16, 17, 18, \dots, 36\} = 15$

3. $(A \cup B) \cap C = \{16, 18, 19, 20, 21, 28, 29, 31, 34\} = 9$

4. $A \oplus B = \{16, 18, 19, 20, 21, 28, 29, 31, 34\} = 9$

C. komposisi fungsi

fungsi

$f(u) = 8u + 7$ 1. $f \circ g = f(g(u)) = 8\left(\frac{u^2}{7} + 1\right) + 7$

$g(u) = \frac{u^2}{7} + 1$

$= \frac{8u^2}{7} + 15$

2. $g \circ f = g(f(u)) = \frac{(8u+7)^2}{7} + 1$
 $= \frac{64u^2}{7} + 16u + 8$

Apakah komutatif

$u = 6 + 2 = 8$

$= f(g(8)) = \frac{8(64)}{7} + 15 = 88.142857 + 15 = 128.142857$
 $g(f(8)) = \frac{64(64)}{7} + 16(64) + 8 = 721.142857 + 1024 + 8 = 1753.142857$

Tidak komutatif karena tidak sama

D. invers

$y = 8u + 7$

$f^{-1}(u) = \frac{u-7}{8}$ $u = k \cdot 100 + a + b = 800 + 7 = 807$

$u = \frac{y-7}{8}$

$u = \frac{807-7}{8} = 100$

e. klasifikasi fungsi

$f(u) = 8u + 7$ = linear gradien $\neq 0$

$g(u) = \frac{u^2}{7} + 1$

injektif

injektif

$= 8u_1 + 7 = 8u_2 + 7 = 8u_1 = 8u_2 \Rightarrow u_1 = u_2$

$= g(2) = g(-2)$, tetapi $2 \neq -2$ X

Maka injektif • Surjektif ✓

Surjektif

Surjektif • Injektif ✓

$u^2 \geq 0 \Rightarrow u^2 \geq 0 \Rightarrow g(u) \geq 1$

$f(u) = y$ • bijektif ✓

Range $g = [1, \infty)$

$8u + 7 = y \Rightarrow u = \frac{y-7}{8}$

Tidak ada u , Maka g tidak surjektif

Maka f surjektif ke $\mathbb{R}$

Maka g bukan bijektif